01 / PURPOSE AND CONTRIBUTION

GlucoTwin: forecasts with inspectable evidence

A working research prototype that forecasts glucose over the next 30 minutes, shows when sensor evidence is unavailable, and records what each prediction actually used.

Why we are building it

A current glucose reading describes the present. We investigate whether recent dynamics improve estimates of the next half hour, and how a review tool should behave when data are missing or delayed. Better-informed review is the intended benefit. Clinical benefit remains unproven.

Team TwinPulse	Program / institution
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What differentiates this implementation

The Forecast Reliability Lab compares original delivery, controlled signal loss and delayed packets on the same recording. Forecast receipts preserve the received-input snapshot and issued path. Recent error uses only target readings that have already arrived. These features make the prototype inspectable; they do not establish a calibrated trust probability or a globally unique invention.

The central measured result

12.64 vs 14.21 mg/dL patient-mean 30-minute MAE: **11.1% lower error** than predicting unchanged glucose, on eight scoreable reserved participants. The test includes only three people with Type 2 diabetes.

This edition supersedes earlier warning counts and submission-report references. No model retraining or threshold tuning occurred during the reliability upgrade. Sources: reports/test_comparison.csv; scoring_v2_audit.json; reliability_benchmark.json.

02 / DATA AND EVALUATION

Linked records, separate participants

Item	Evidence
Dataset	CGMacros v1.0.0, PhysioNet. 45 linked participants: 14 Type 2, 16 prediabetes, 15 without diabetes.
Retrieval	48 numeric CSV members, member CRC32 checks and SHA256 records. Full photo archive not retrieved or checksum-verified.
Clinical fields	Age, BMI, HbA1c and fasting glucose.
Dynamic fields	CGM with available heart rate and activity. Meals, Libre, medication guesses and participant IDs are excluded features.
Partitions	27 training / 9 validation / 9 reserved test. No participant crosses partitions.
Scoring sample	18,746 common forecast timestamps, 8 scoreable patients. One unsupported test patient remains in coverage.

Preprocessing caveat

The source minute-level stream contains interpolation. We infer a fixed five-minute phase from integer CGM values in the first 24 hours, reject fractional/out-of-range endpoints, and do not fill gaps. This cannot fully recover original device timestamps or hidden source interpolation. Replication on native CGM observations remains necessary.

How selection stayed separate

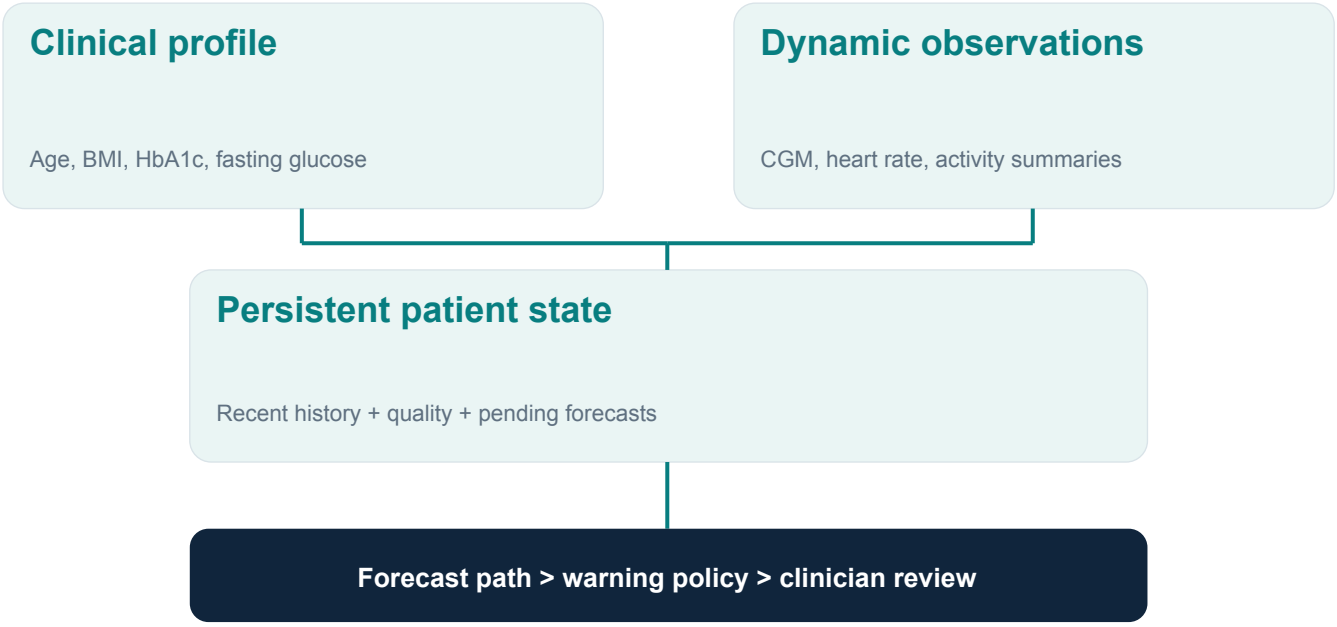
Validation selected the model, residual intervals, warning thresholds and bias-update rate before the recorded test. The new disruption study uses already inspected reserved recordings and is an exploratory engineering stress test. It provides no new independent validation of accuracy.

Population and license limits

The cohort was recruited in the US. No Indian-population or clinical-benefit validation exists. Original code uses MIT. CGMacros data use CC BY-NC-SA 4.0 and are separately attributed. Models and raw records remain local.

03 / ARCHITECTURE

A changing state with a forecast ledger



The patient state receives thirteen consecutive five-minute observations spanning an hour. Each update matures old forecast errors, checks the input history and emits estimates at +5 through +30 minutes. Missing required history produces a withheld state.

The evidence ledger

ForecastLedger records the model artifact hash, clinical-profile usage, received observations, issuance time and six target values. A SHA256 digest covers the fixed issued payload. Later outcomes append separately when actual target readings arrive. Missing targets remain unobserved and never become zero error.

Controlled delivery checks

The lab preserves the source timestamp of delayed packets. Ten-minute-old packets fail the five-minute freshness rule. The current input becomes unavailable, with no timestamp relabeling, carry-forward or retroactive backfill. Forecasts resume after thirteen fresh consecutive observations.

Clinical fusion engines use both clinical and dynamic fields. Validation selected the CGM-only engine and a bias-update rate of zero. The twin is a data-driven patient-state representation, with no validated physiological reconstruction or learned personalization benefit.

04 / FORECAST RESULTS

11.1% lower average 30-minute error

Model	Validation MAE	Test MAE	Type 2 test
Unchanged glucose	11.37	14.21	16.43
Recent trend	16.72	19.80	20.30
Linear CGM	10.79	13.19	14.51
Boosted CGM - selected	10.55	12.64	14.12
Clinical + CGM	10.65	12.53	14.03
Clinical + CGM + activity	10.62	12.47	14.04

Patient-mean MAE in mg/dL at +30 minutes. Test: eight scoreable participants; Type 2 subset: three. All models use the same scored timestamps. Forecast metrics are unchanged by the warning-scoring correction.



Patient-mean 30-minute MAE, mg/dL | Lower is better

The selected model reduces overall MAE by 11.1%, and Type 2 subgroup MAE by about 14.1% versus unchanged glucose. Test RMSE remains 18.62 versus 21.14 mg/dL. Lower mean error does not eliminate large mistakes.

Clinical fusion is implemented, but its benefit remains uncertain

Fusion is slightly better numerically on this test, but did not win validation. We retain the validation winner. Bias correction rates above zero worsened validation and remain disabled.

05 / CORRECTED WARNING ACCOUNTING

Causal emission and explicit censoring

A research onset crosses strictly above 180 mg/dL and remains above across four five-minute observations. The selected model warns when its forecast path reaches 170 mg/dL, while the current reading is at most 180. Emission uses only present inputs, with a 30-minute cooldown.

Scoring v2 / selected model	All cohorts	Type 2
Credited / eligible onsets	162 / 163	106 / 106
Missed onsets	1	0
False warnings	268	161
False warnings / eligible non-event hour	0.232	0.466
Censored emitted warnings	3	1
Median credited lead	10 min	10 min

What the correction changes

Scoring v1 suppressed all alert opportunities unless 45 minutes of future readings were complete. It excluded a valid advance forecast at P049 before a 07:33 onset because a gap began at 07:58. Scoring v2 emits and advances cooldown causally, credits fully observed positive onset evidence, and censors unmatched alerts without full follow-up. Three emitted alerts have unknown outcomes and do not count as false warnings.

Frozen models, input forecasts and thresholds stay unchanged. Persistence now credits 139/163 onsets at 0.213 false warnings/hour. The selected model credits more events with more false warnings. P041 remains a genuine missed recrossing. Old outputs are preserved in reports/scoring_v1/.

Interpretation

These are corrected retrospective research counts, not overall accuracy or prospective clinical validation. Thresholds were originally selected with scoring v1 and have not been re-optimized. False-warning rates describe eligible observable non-event time; missing follow-up remains an evidence limit.

06 / RELIABILITY BENCHMARK

The measurable cost of disrupted data

The frozen selected model runs under three controlled delivery schedules across all nine reserved participants. The schedule uses elapsed time, never outcome labels. Fifteen minutes of loss or thirty minutes of delayed delivery recur every six hours, beginning at the three-hour phase after initialization.

Scenario	Coverage	MAE	Normal MAE on same targets
Original delivery	78.8%	12.64	12.64
15-min signal loss / 6h	62.4%	12.60	12.60
10-min delay for 30 min / 6h	59.2%	12.59	12.59

Coverage is the patient mean across all nine reserved recordings, including the unsupported participant. MAE averages the eight scoreable participants on observed targets retained in each scenario. Units: mg/dL at +30 minutes.

What this shows

Controlled loss withholds 3,909 additional grid opportunities; delayed delivery withholds 4,680 and rejects 2,213 stale packets. Forecast curves overlap wherever the received input windows are unchanged. The offline benchmark reuses the identical frozen forecasts after applying the causal support mask, with incremental parity checks for the delivery logic.

A lower retained-set MAE is not better prediction

Each disruption removes some targets from scoring. The paired normal comparison uses those same retained targets and matches exactly. The demonstrated behavior is withholding and recovery, with reduced availability. We do not claim that rejecting input improves predictive accuracy.

What the lab does not establish

The scenarios model transport and availability failures, not glucose physiology, real hardware reliability or clinical safety. Different fault schedules, native device records and independent participants could produce different results. Protocol and all per-patient results are in reports/reliability_benchmark.json.

07 / DEMO AND VERIFICATION

How to inspect and reproduce the evidence

A demonstration judges can verify

Open Reliability lab, choose an anonymous participant and compare the same 6/12/24-hour segment under all three delivery conditions. Inspect the delivery table for source timestamps and stale rejection. Download the scenario receipt JSON, then inspect an input snapshot, its issued path and the subsequently observed outcomes. The receipt digest checks content consistency; it is not a signature or authenticated medical record.

Recent performance has an honest denominator

Patient twin shows the MAE of the most recent twenty observed matured +30-minute forecasts. Before a target arrives, its error is unavailable. Missing targets stay excluded. These overlapping local errors describe the replay history and are not a calibrated trust score.

Verification and reproduction

Tests cover future isolation, withheld-state recovery, receipt immutability/hash verification, delayed-source timestamps, missing targets, retained forecast parity, positive onsets followed by gaps, censored-alert cooldown, trained-model incremental equivalence, and dashboard interactions. The frozen forecast/evaluation verification reproduces the updated reports.

Commands from the glucotwin directory

```
python3 scripts/check.py
python3 train.py final --verify
python3 scripts/rescore_events.py
python3 scripts/benchmark_reliability.py
python3 run.py
```

Code: core.py, reliability.py, evaluation.py, app.py. Evidence: reports/scoring_v2_audit.json; reliability_benchmark.json; test_results.json; test_comparison.csv. Dataset: CGMacros v1.0.0, PhysioNet, DOI 10.13026/3z8q-x658. Dataset citation: Gutierrez-Osuna, Kerr, Mortazavi & Das (2025).

The project is a research review tool with no treatment recommendations or demonstrated health benefit. Public repository and video accessibility must be checked separately before submission.